

Ananya Appan

RESEARCH INTERESTS	Cryptography and Distributed Computing: Secure Multi Party Computation, Verifiable Secret Sharing, Consensus, Oblivious RAM	
EDUCATION	<i>Ph.D (3rd year)</i> University of Illinois Urbana Champaign , • CGPA : 4 / 4 • Supervisors : David Heath, Ling Ren	Aug 2023 - Present
	<i>Integrated Master of Technology</i> International Institute of Information Technology (IIIT) Bangalore , India • CGPA : 3.84 / 4 • Supervisor : Ashish Choudhury	July 2022
	<i>Intermediate (12th Class)</i> Chetana Pre University College, Bangalore, India • Karnataka Pre University Board • Percentage : 98%	May 2017
	<i>Secondary (10th Class)</i> Delhi Public School Bangalore North • CBSE (Central Board of Secondary Education) • CGPA : 10 / 10	March 2015
JOURNAL PUBLICATIONS	• Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, Perfectly-Secure Synchronous MPC with Asynchronous Fallback Guarantees, <i>IEEE Transactions on Information Theory</i>, 2023 • Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, Revisiting the Efficiency of Asynchronous MPC with Optimal Resilience Against General Adversaries, <i>Journal of Cryptology</i>, 2023	
CONFERENCE PUBLICATIONS	• Ananya Appan, David Heath, Ling Ren Verifiable Secret Sharing, Randomized Agreement, and Multi-Party Computation in Granular Synchrony : <i>ASIACRYPT</i> 2025 • Ananya Appan, David Heath, Ling Ren Oblivious Single Access Machines: A New Model for Oblivious Computing : <i>ACM Conference on Computer and Communications Security (CCS)</i> 2024 • Ananya Appan, Ashish Choudhury, Network Agnostic MPC with Statistical Security : <i>Theory of Cryptology Conference (TCC)</i> 2023 • Ananya Appan, Ashish Choudhury, Perfectly Secure Synchronous MPC with Asynchronous Fallback Guarantees Against General Adversaries : <i>International Symposium on Distributed Computing (DISC)</i> 2023 • Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, Perfectly-Secure Synchronous MPC with Asynchronous Fallback Guarantees : <i>ACM Symposium on Principles of Distributed Computing (PODC)</i> 2022	

UNPUBLISHED MANUSCRIPTS

- Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, [Revisiting the Efficiency of Asynchronous Multi Party Computation Against General Adversaries](#) : *INDOCRYPT 2022: 23rd International Conference on Cryptology*
- Ananya Appan, David Heath: [Oblivious Immutable Memory](#)
- Sage Pia, Ananya Appan, Maryam Rezapour, Amey Shukla, Nikhil Date, Benjamin Fuller, Ling Ren, David Heath: [Oblivious Single Access Machines are Concretely Efficient](#)
- Pranav Shriram Arunachalam, Ananya Appan, David Heath, Ling Ren: [Efficient Private Range Queries on Public Data](#)
- Ananya Appan, David Heath, Ling Ren: [Asynchronous MPC with Abort](#)

TALKS

- Invited talk at [University of Connecticut](#) Security Seminar Nov 2024
- Based on the paper “Oblivious Single Access Machines”
- [Bangalore Crypto Day, Spring 2023](#) March 2023
- Based on the paper “Revisiting the Efficiency of Asynchronous Multi Party Computation Against General Adversaries” at the Indian Institute of Science.
- [Theory and Practice of Multi Party Computation \(TPMPC\) 2022](#) June 2022
- Remote talk based on the paper “Perfectly-Secure Synchronous MPC with Asynchronous Fallback Guarantees” ([recording](#)).

TEACHING ASSISTANT- SHIPS

- University of Illinois, Urbana Champaign*
- Secure Computation August 2025 - December 2025.
- Instructors : Dr. [David Heath](#)
- International Institute of Information Technology, Bangalore*
- Conducted (online) tutorial sessions, set questions for assignments and exams, and evaluated these for the following courses.
- Foundations of Cryptography January 2022 - May 2022.
Instructors : Dr. [Ashish Choudhury](#), Dr. [Srinivas Vivek](#)
 - Discrete Mathematics August 2021 - December 2021.
Instructor : Dr. [Ashish Choudhury](#)
 - Data Structures and Algorithms March 2021 - July 2021.
Instructor : Dr. [V N Muralidhara](#)
 - Programming II (C++ and Java) August 2020 - December 2020.
Instructors : Dr. [Jaya Sreevalsan Nair](#), Dr. [T K Srikanth](#)
- [National Programme on Technology Enhanced Learning \(NPTEL\)](#)
- Instructor : Dr. [Ashish Choudhury](#)
- Set questions for assignments and exams, and helped manage the discussion forum for the following MOOC (Massive Open Online Courses) courses.
- [Secure Computation: Part II](#) July 2022 - November 2022.
 - [Discrete Mathematics](#) January 2022 - April 2022.
 - [Foundations of Cryptography](#) January 2022 - April 2022.
 - [Secure Computation: Part I](#) July 2021 - November 2021.

WORK EXPERIENCE

Associate Developer

July 2022 - Present

[SAP Labs, India](#)

- Build cloud applications on SAP's Business Technology Platform ([BTP](#)) using SAP Cloud Application Programming ([CAP](#)) model.

Summer Intern

May 2021 - July 2021

[SAP Labs, India](#)

- Created dashboards using SAP Analytics Cloud ([SAC](#)) and worked on a proof of concept for querying encrypted databases.

Research Intern

May 2020 - July 2020

[Indian Institute of Science, Bangalore](#)

- Supervisor : Dr. [Jayant R. Haritsa](#)
- Compared [UNMASQUE](#), a hidden SQL query extractor, to other implementations for reverse query engineering, and built a synthetic SQL query generator for testing it.

Developer

June 2018 - December 2018

[LexHeal, Bangalore](#)

- Helped build the front end of an EHR Mobile Application for a start up using React Native.

ACHIEVE- MENTS

- Received the Late Sri. N. Rama Rao Gold Medal for [Student of the Year](#) for the graduating batch of 2022.
- Included in the [Dean's Merit List](#) of IIIT Bangalore from 2017 to 2022.
- Selected for being awarded the [IAS](#) (Indian Academy of Sciences) fellowship under the Computer Science Engineering category in 2020.
- Finalist in [LinkedIn Wintathon 2020](#).
- Part of the team which received second place in the [Codess](#) Hackathon conducted by Microsoft in 2019.
- Qualified for [ACM ICPC](#) Regionals (Amritapuri and Kharagpur) in 2018.
- Among top 1000 students in the country selected for being awarded the [KV PY](#) Scholarship in 2016.
- Ranked 24th in the state in the NTSE (National Talent Search Exam) in 2014.